



Assessment of groundwater potential zones through remote sensing and electrical resistivity surveys: A case study of Peddavanka watershed, Cuddapah Basin, India

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Abstract

Peddavanka watershed comprising quartzite, shale, limestone, and basic intrusives (sills) as the major rock formations of the Pre-Cambrian age group. The watershed lies in between Latitudes 14°55'24"-15°12'50" N and Longitudes 77°51'20"-78°03'26" E, in Anantapur District, Andhra Pradesh, India. The objective of the present study is to assess the resources of groundwater potential through hydrogeomorphological and electrical resistivity surveys. Hydrogeomorphological map and lineament density map have been prepared based on the visual interpretation of satellite Geocoded data of IRS-1B, LISS-II (57 E/16, F/13, J/1 and I/4) on 1:50,000 scale. Vertical electrical resistivity soundings were conducted covering the entire watershed by adopting Schlumberger configuration of the surface geophysical method to delineate the groundwater potential aquifers in the watershed. The data were interpreted with the help of master curves and auxiliary point charts. Interpretations of VES were used to generate a top layer apparent resistivity

contour map and longitudinal conductance map. Isoresistivity contour maps were prepared and interpreted in terms of resistivity and thickness of various sub-surface layers, and isocontour diagrams depicting the depth to bedrock were prepared. Resistivity results were correlated with the existing lithology. Based on the depth to bedrock, the thickness of the saturated layer and the resistivity of the second layer, a groundwater potential map has been prepared, in which good, moderate, and poor zones are classified. Remote sensing and electrical resistivity studies have been correlated. Both the studies clearly reveal that the flood plains, alluvial plains, and valley fills with thick alluvium, weathered and fractured portions of shale and limestone lying in southern and central portions of the watershed constitute the productive water-bearing zones categorized as good groundwater potential aquifers.

Keywords

Electrical resistivity –
Hydrogeomorphology – Lineament
Density – Groundwater potential
aquifers – India

1. Introduction

Peddavanka, a tributary to river Pennar, rises from Mutchukota hills of Erramalai hill ranges flows southerly and joins the Pennar river near Tadipatri of Anantapur District, Andhra Pradesh, India. It includes the sub basins of Neredu vanka, Nittur vanka, Gudipadu vanka, and Bugga vanka. This originates in the southern part of Anantapur district and drains through northeastern part of Anantapur district covering an area of about 298 km², located in the Survey of India Toposheet numbers 57 E/16, F/13, J/1, and I/4 on a scale of 1:50,000 scale. The study area lies between Latitudes 14°55'24" – 15°12'50" N and Longitudes 77°51'20" – 78°03'26" E (Fig.1).



Fig.1. Location map of Peddavanka watershed.

2. Geology

Most of the area under watershed is occupied by Tadipatri formation of Lower Cuddapah Super group

which is mainly represented by shale with interbedded layers of quartzite and flaggy limestone. The shale trends NNE-SSW with about 10° dipping towards east. Shale often contains layers of basic volcanic rocks including sills of trap and trap conglomerate with volcanic pellets. Tadipatri formation is overlain by the rocks of the Kurnool group, which include Banaganapalle quartzite, Narji limestone and Auk shale. Alluvium of recent age is comprised of sand, silt, and clay occurring all along the course of the stream and at the confluence of the stream with Pennar river in the southern portion of the watershed (Fig.2).

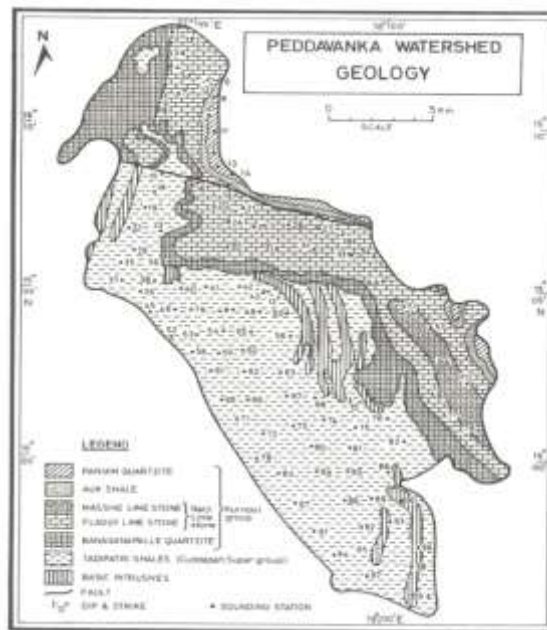


Fig.2. Geological map along with VES locations.

3. Hydrogeomorphology and Remote Sensing

The systematic study of various landforms and the geomorphic units has been carried out in the

watershed area. Hydrogeomorphological map is prepared based on the visual interpretation of Satellite Geocoded data of IRS-1B, LISS-II (57 E/16, I/4, F/13, J/1) on 1:50,000 scale (Fig.3). The data is correlated with the Survey of India topographic maps on 1:50,000 scale and by conducting field checks. The landforms identified in the watershed area are valley fills, flood plains and alluvial plains deposited by the fluvial processes.

3.1 Valley Fills (VF)

The unconsolidated materials partly filling a valley formed by the depositional processes at the youthful stage of a river consisting of pebbles, gravel, sand, and silt brought by the streams. They show high range of grain size and less amount of compaction expressing high permeability resulting in high infiltration. Owing to large amounts of recharge they are considered as favourable groundwater zones. These valley fills are identified in the study area along the river course.



Fig.3. Hydrogeomorphological map

3.2 Flood Plain (FP)

Flood plains are the landforms formed by dumping of sediments along the river banks during the floods. Flood plains are composed by unconsolidated and loose sediments like gravel, sand and silt found along the banks of river course. A good amount of moisture is retained, hence facilitates more vegetal cover and agriculture in the flood plains. From the satellite imagery interpretations, flood plains are found characteristically associated with the confluence of Peddavanka with Pennar river. The groundwater prospects are excellent in the flood plain areas.

3.3 Alluvial Plain (AP)

Alluvial plains are level plains or gently sloping plains or slightly undulating land surface comprising alluvial deposits of sand, silt, and clay of varying lithology built on either side of a river. Generally the groundwater prospects are high in these areas. These alluvial plains are found on the western side of the river course extending from north to south in the watershed area.

3.4 Denudational Landforms

The denudational landforms are found as a result of active processes of weathering, mass wasting, and erosion caused mainly by running water. The landforms formed by the agents of denudation are identified from the satellite imagery are residual hills, moderately weathered, shallow weathered pediplains which are moderate to poor groundwater potential zones.

3.5 Structural Landforms

Various lithology including sedimentary and metamorphic rocks in combination with a variety of geological processes produced a great variety of characteristic structural landforms in the study area. The lithologic aspects such as texture, porosity and structural elements such as dip, strike, fractures, fault, fold etc. control the formation of the structural landforms. Various structural landforms present in the study area are Mesa/Butte, Cuesta hills and structural hills composed of Paniam Quartzite, Auk Shale, Narji Limestone, and Tadipatri shales.

3.6 Lineaments

A number of mega and micro lineaments are identified from the satellite imagery. Many lineaments are trending NE-SW direction roughly parallel with the trends of different geological formations while the rest are running in NW-SE direction. There are a few lineaments that align nearly North-South direction. A major fault is identified on the northern part of the basin running NNW-SSE cutting across about all the geological formations present in the basin with brecciation, small fractures, and jointing as evidence faulting. The fault is observed to extend far beyond the boundaries of the watershed on either side. Similarly the lineaments trending NNE-SSW in the central portion of the watershed nearly match with the alignments of the stream courses of the watershed. Thus the Peddavanka drainage in general and sub-basins in particular can be considered as controlled by fracture.

4.1 Lineament Density

Based on the concentration and length of lineaments, the lineament density map has been prepared (Fig.4). Secondary porosity is imparted by joints and fractures in the areas of higher values of lineament density. The lineament density map reveals the variations of groundwater potentiality in the watershed. The high lineament density with more than 2 is noticed as isolated patches mostly along the river course indicating high groundwater potential especially associated with Tadipatri shales. The medium lineament density ranging from 1 to 2 indicates the moderate groundwater potential noticed in areas along the river course and in other areas of the watershed. The low lineament density of less than 1 is occupied by large areas in the watershed indicating poor groundwater potential. Based on the lineament density it is inferred that most of the area under watershed is of low groundwater potential.



Fig. 4 Lineament density map.



4. Geomorphology and Groundwater

The hydrogeomorphological map is prepared from the satellite imagery is helpful in identifying favourable zones of groundwater. The various geomorphic units are classified as favourable, moderately favourable, and poor zones of groundwater. Since flood plains, alluvial plains, and valley fills are associated with thick alluvium and weathered material, groundwater recharge is generally high in these areas and hence these areas are marked as favourable for groundwater exploitation. Only a small area is occupied by alluvial and flood plains in the watershed while the valley fills are distributed in the northern side. Moderately weathered and shallow weathered pediplains of shale formations are distributed all along the river course. As these geomorphic units indicate low permeability characteristics they can be classified as moderately favourable zones of groundwater. As mesas / buttes and cuestas of Paniam and Banaganapalli quartzite and Tadipatri and Auk shale and Narji limestone do not favor much infiltration, hence they are considered as poor zones for groundwater exploitation.

5. Electrical Resistivity Survey

Ninety-nine vertical electrical soundings, randomly distributed, were conducted in the watershed, by adopting Schlumberger configuration and their locations are given in Fig. 1. The VES data are interpreted using the curve matching technique with the help of master curves prepared by Rijkwaterstaat (1969), and the results and salient features of the subsurface parameters. The sounding curves A, K,

and H type suggest a few three-layer geoelectrical sections, and a number of four-layer sections of KH, HK, QH, HA, AK, and AA types. The types of curves obtained are predominantly A-type followed by KH-type, H-type, K-type, QH-type, HA-type, HK-type, and AA-type. Iso-resistivity contour maps and the bed rock contour maps, as well as their 3-D maps, were prepared using the SURFER package, and they were interpreted in terms of resistivities and thicknesses of various subsurface layers. Based on the VES interpretation results, the top layer contour map, longitudinal conductance map, depth to bedrock map, and the groundwater potential map are prepared.

5.1 Correlation of results of soundings with lithology

The results from four resistivity probes conducted in close proximity to four wells in different formations have been correlated with the known lithology, as shown in Fig. 5. Sounding nos. 1 and 31 were in the limestone terrain, while sounding nos. 41 and 95 were in shale. The results from sounding no. 1 indicate four layers with absolute resistivity of 23, 897, and 228 Ω m, respectively, for the first three layers, and the bottommost layer has very high resistivity. The depth of the well is 55 m, and the first layer with 23 Ω m correlates to the top soil, which extends down to a depth of 1.7 m from the surface. Between 1.7–23.8 m the resistivity increases to 897 Ω m, indicating massive limestone. The third layer from 23.8–30.5 m shows a decreasing trend in resistivity, indicating fractured limestone, which

yields significant quantities of groundwater. Similarly, the results from sounding no. 31 conducted at Kundanakota village indicate four layers, with the top soil exhibiting a resistivity of $11 \Omega\text{m}$ up to a depth of 2.1 m, the second layer with a resistivity of $33 \Omega\text{m}$, indicating clay from 2.1–6.3 m, and the third layer, a fractured limestone with a resistivity of $131 \Omega\text{m}$ that occurs between 6.3–60.7 m. The well was drilled up to a depth of 65 m, yielding significant quantities of groundwater. The dense limestone at the bottom is the fourth layer with high resistivity. Limestone, though very compact, is known to be a good water-yielding formation, due to its secondary porosity such as joints, fractures, bedding planes, and also due to being cavernous. The results from the analysis of sounding nos. 41 and 95, at Kamalapadu and Kaverisamudram villages, respectively, suggest that their well log data corroborates perfectly with the lithology. The top soil with clay is the first layer in both borewell logs, up to a maximum of 1.7 m. Shale with red clay to a maximum of 6.3 m, exhibiting resistivity of $24 \Omega\text{m}$ and $31 \Omega\text{m}$, is the second layer. Weathered shale, highly variable in thickness from 16–38.8 m, with resistivity of $15\text{--}16 \Omega\text{m}$, is identified as the third layer. The total depth of the bore wells was 30 m and 42 m at sounding nos. 41 and 95, respectively. The bottom-most layer, with resistivity of $141 \Omega\text{m}$ and $226 \Omega\text{m}$, is hard shale.

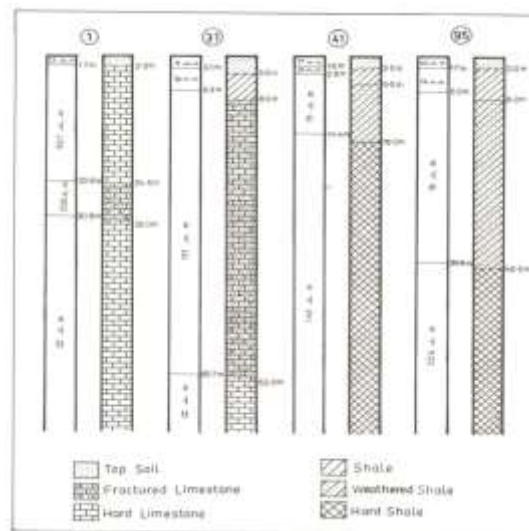


Fig.5. Correlation of resistivity results with bore-hole data

5.2 Groundwater Potential Aquifers

Assuming that wide variations are not present within a few kilometres, groundwater potential aquifers (Fig. 6), which are classified as good, moderate and poor zones, have been delineated by interpreting the depth to bedrock, the thickness of the saturated layer, and the resistivity of the second layer. Limestones and shales, which are devoid of primary porosity, often act as the potential aquifers when they are fractured and/or weathered (Singh and others 2003). The study reveals that the weathered and fractured portions in shale and cavernous limestone that occur in the southernmost and central portions of the watershed area contain productive water bearing zones categorized as good potential zones. Moderate groundwater potential zones occur mainly in the western and central portions, and subordinately in the northern and southern portions of the basin, which

are occupied by shale and limestone. The remaining parts of the area were demarcated as the very low saturated zone, classified as a poor potential water zone in the dense and massive limestone and in shale.

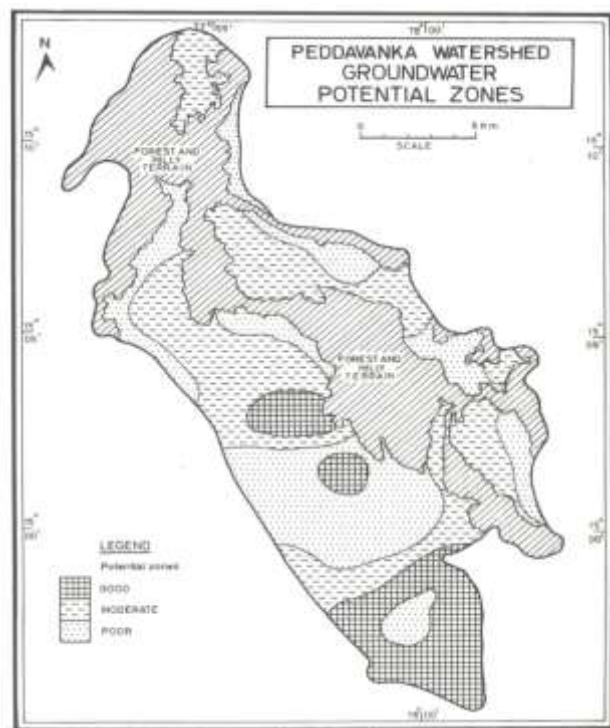


Fig.6. Ground water potential zones Map

6. Conclusions

Through investigations conducted by the electrical resistivity survey, groundwater potential aquifers have been delineated. Fractured and weathered horizons have been identified in the study area underlying VES stations, and all of these constitute good aquifer zones. Good prospects therefore exist for groundwater development in the study area where the depth to basement is relatively thick and has favourable low resistivity values, while those with

thin depth to bedrock and high resistivity value have a lower potential for an aquifer. The interpretation of depth to bedrock is in good agreement with the actual depth observed in the dug wells, dug-cum-bore wells and bore wells. The limestone, owing to its cavernous nature, produces high yields that occur as springs at a few locations in the watershed on the one hand, while it is very hard and massive without any weathered portion and is totally devoid of water at some places on the other hand. The electrical resistivity data therefore gives reasonably accurate results that can be used to understand the subsurface layers and basement configuration in groundwater exploration.

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